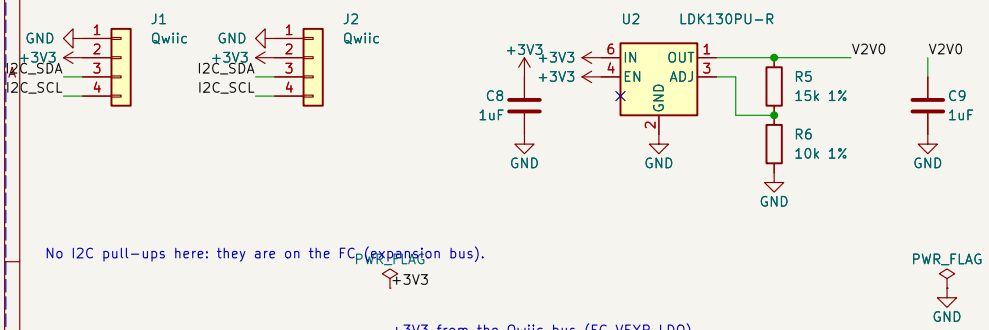
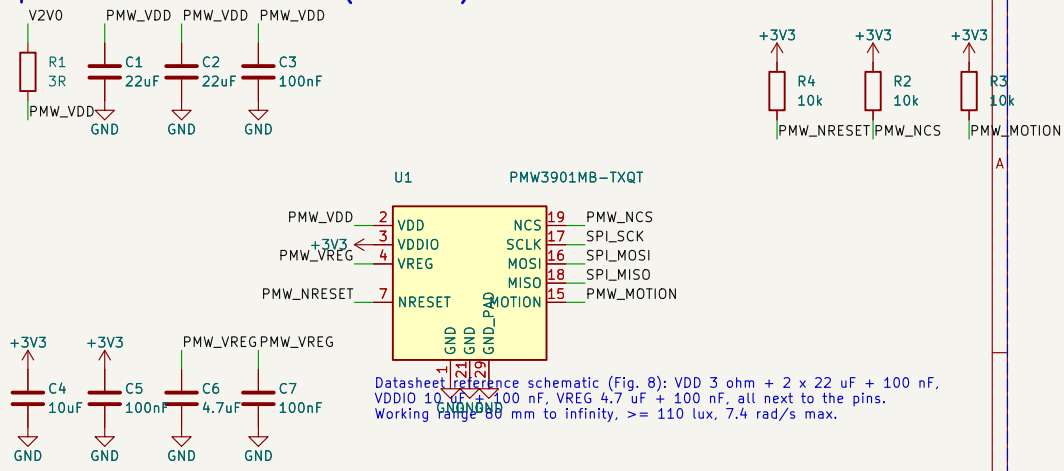
A — Qwiic in/out, 2.0 V for the flow sensor



No I2C pull-ups here: they are on the FC (expansion bus).

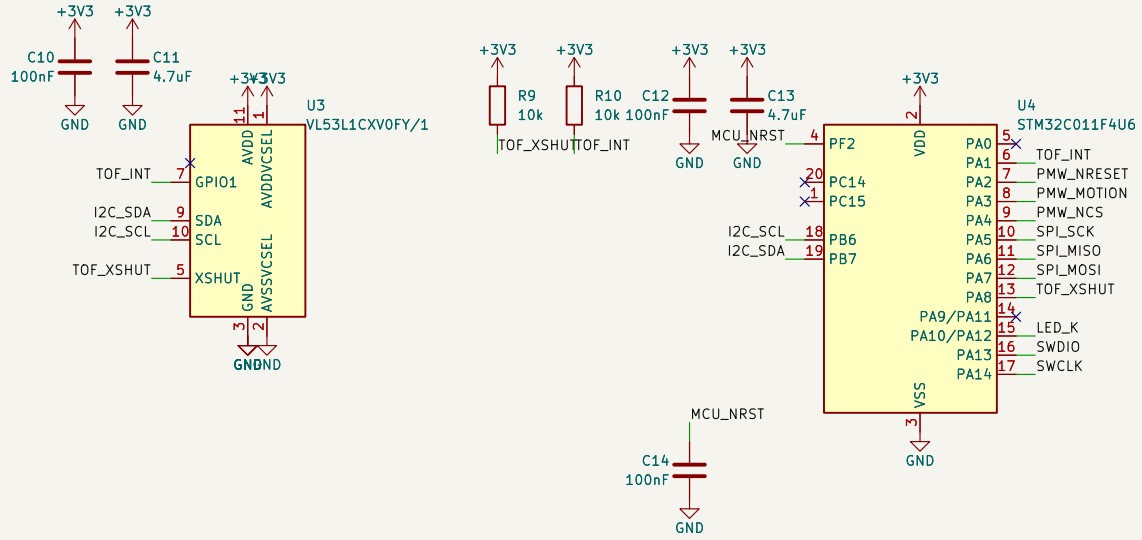
+3V3 from the Qwiic bus (FC VEXP LDO).
Total draw about 35 mA (flow 9 mA, ToF up to 20 mA, MCU 3 mA).

B — Optical flow PixArt PMW3901MB (SPI 2 MHz)



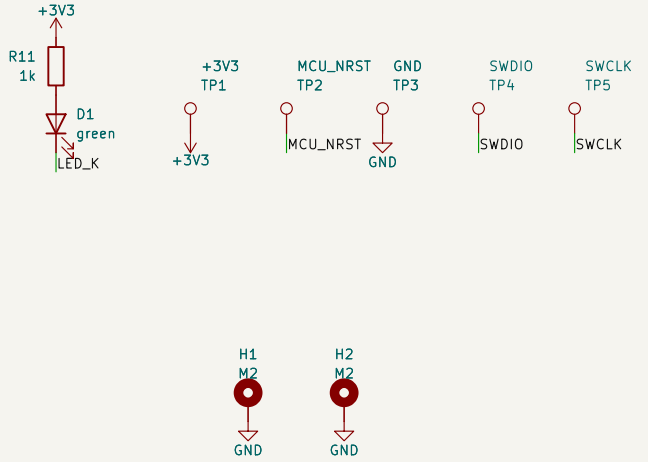
Datasheet reference schematic (Fig. 8): VDD 3 ohm + 2 x 22 uF + 100 nF, VDDIO 10 nF + 100 nF, VREG 4.7 uF + 100 nF, all next to the pins.
Working range 80 mm to infinity, >= 110 lux, 7.4 rad/s max.

C — ToF ST VL53L1X (bus 0x29) and bridge MCU ST STM32C011 (bus 0x3B)



Bus map: ToF 0x29 (VL53L1X default), flow bridge 0x3B.
The bridge can hold the ToF in reset (XSHUT) to change its address.

D — Status LED, SWD pads, mounting



Generated automatically: run ERC before release
PixArt PMW3901MB (flow, SPI) behind an ST STM32C011 I2C bridge; ST VL53L1X ToF on the bus
Qwiic (JST SH 4P) in/out: daisy-chain with the GPS board; I2C 0x29 (ToF) and 0x3B (flow bridge)
Mounted under the battery tray, sensor side facing the ground (handoff doc for the frame)

ACwing — Andrea Cirillo

Sheet: /
File: ACwing_Flow_A.kicad_sch

Title: ACwing Flow rev A — optical flow + ToF expansion for ACwing FC rev A

Size: A4	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/1